

Bimetallic Plasmonic Nanoparticle Lattices for Photocatalytic Chemical Transformations

Guanyu Lu, Nam Heon Cho, Sachin P. Kulkarni, Matthew Hershey, Xin Wen, Sang-Min Park, Koray Aydin, Teri W. Odom,* and Dayne F. Swearer*



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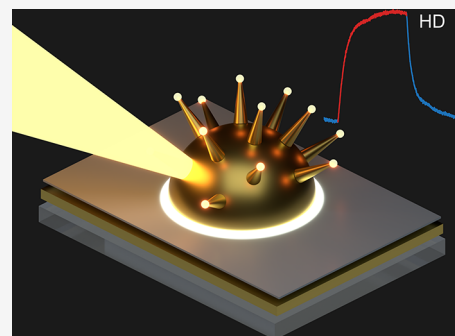
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ABSTRACT: Plasmonic metasurfaces offer unique opportunities to manipulate light at the nanoscale and drive photocatalytic reactions, but fabrication challenges and inefficient catalytic interfaces have limited their use in heterogeneous catalysis. Here, we report scalable bimetallic metasurfaces, fabricated by combining soft lithography with templated chemical synthesis, that consist of centimeter-scale arrays of spiky Au nanoparticles decorated with Pt (spiky Au@Pt). These arrays enhance the rate of the hydrogen–deuterium exchange reaction ($\text{H}_2 + \text{D}_2 \rightleftharpoons 2\text{HD}$), a benchmark photocatalytic process, by 13-fold compared to smooth Au@Pt arrays and reduce the apparent activation energy by nearly half. This activity also exceeds that of similar planar photocatalyst systems reported previously, even at lower illumination intensities. Wavelength-dependent activity, electromagnetic simulations, and quantum-based calculations reveal that strong near-field confinement at sharp Au tips and resonant gap-plasmon excitation promote plasmon-enhanced reactivity, highlighting nanoscale morphology engineering as a pathway to improve photocatalytic metasurfaces.

KEYWORDS: Plasmonic metasurfaces, heterogeneous photocatalysis, bimetallic nanoparticles, plasmon-induced chemistry, antenna-reactor nanostructures



Recent breakthroughs using metallic nanoparticles (NPs) with localized surface plasmon resonances have unveiled emerging pathways for transforming electromagnetic energy into chemical reactions^{1–4} via strong light-matter interactions.^{5,6} Although plasmonic nanostructures exhibit enhanced near-fields,^{7,8} highly energetic carriers,^{9,10} and localized photothermal effects,^{11,12} their practical catalytic efficiency is often limited by the inherently low activity of plasmonic metals when used alone.¹³ To address this limitation, colloidal plasmonic NPs (e.g., Au, Ag, Cu, Al)¹⁴ can be combined with catalytic metals arranged as islands or dispersed single atoms,^{15,16} forming “antenna-reactor” structures^{17–19} that achieve high efficiency and light-dependent selectivity.^{20–22} However, the optical response of these colloidal systems remains constrained by NP size and intrinsic material properties, restricting energy tunability.²³ Additionally, random spatial distributions and aggregation yield inhomogeneous optical properties,²⁴ while dense catalyst loading can limit visible light penetration and energy delivery throughout a packed bed.²⁵

As an alternative, plasmonic metasurfaces²⁶ composed of well-defined subwavelength nanostructures could decouple the intrinsic optical properties of individual metal NPs,²⁷ offering greater control over hot-carrier generation and photothermal effects by design compared to supported NP counterparts.²⁸ Additionally, active layers in planar metasurfaces are thin

(usually sub- μm for optical metasurfaces),^{29,30} mitigating the issue of limited light penetration in packed catalyst beds.³¹ Thus, plasmonic metasurfaces provide a promising platform for manipulating light at the nanoscale in photocatalytic applications, and their highly customizable optical properties could enable small-scale, decentralized photochemical processes.²⁵

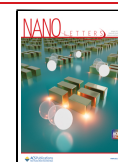
Despite this possibility, plasmonic metasurfaces remain underexplored in photocatalytic applications, with only a few examples reported to date.^{31–33} A key barrier is that conventional fabrication methods,^{28,34} such as electron-beam lithography, have inherently high infrastructure costs and face significant challenges when scaling to large-area arrays, hindering their broader adoption in sustainable photocatalysis.^{25,35} Alternatively, soft-lithography-based methods offer a cost-effective route to large-scale fabrication of plasmonic metasurfaces through simple mask-to-substrate transfer.³⁴ Techniques such as solvent-assisted nanoscale embossing (SANE)³⁶ and phase-shifting photolithography

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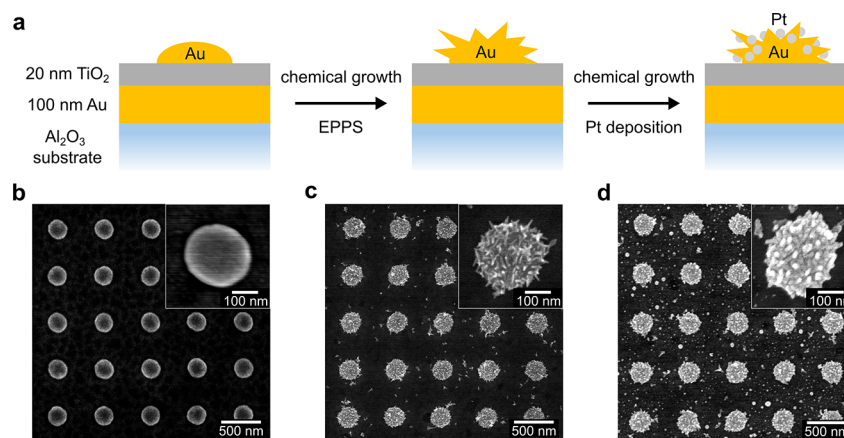


Figure 1. Fabrication of spiky Au@Pt. (a) Diagram representing the fabrication procedure for spiky Au@Pt. First, the thermally annealed Au NPs array is prepared by the SANE technique on the multilayer substrate consisting of a 20 nm TiO_2 layer on a 100 nm Au layer on the Al_2O_3 substrate. Then, the spiky Au NPs are formed through a chemical growth procedure in EPPS buffer. Finally, Pt NPs are deposited via chemical growth. (b) SEM image of Au NPs array with the inset showing one individual Au NP. (c) SEM image of spiky Au NPs array with the inset showing one individual spiky Au NP. (d) SEM image of spiky Au@Pt NP array with the inset showing one individual Au NP.

(PSP),³⁷ coupled with reactive-ion etching (RIE),³⁴ yield uniform and periodic plasmonic NP lattices for efficient light-matter interactions.^{38–42} Moreover, these lattices can serve as versatile templates for synthesizing more complex nanomaterials,⁴³ enabling the fabrication of bimetallic NP arrays^{44,45} and morphology-engineered NP arrays.⁴⁶ For example, chemical incubation using Good's buffers⁴⁷ facilitates anisotropic NP growth, producing spiky Au NP arrays with strongly enhanced local electric fields at high-curvature tips,⁴⁶ which could increase photocatalytic activity. Despite these advances, such scalable, morphology-engineered plasmonic NP lattices have yet to be fully exploited to address limitations in photocatalytic efficiency and energy delivery.

In this work, we address the challenges associated with scalable plasmonic metasurface fabrication and limited catalytic performance. We designed bimetallic plasmonic lattices featuring sub-10 nm protrusions surrounding periodic Au NPs decorated with Pt (spiky Au@Pt). Using a combination of soft-lithography-based patterning and solution-phase chemical growth performed directly on the patterned arrays, we fabricated centimeter-scale planar structures for photocatalytic studies. Such spiky Au@Pt exhibits 13-fold higher photocatalytic rates and reduced apparent activation energies, as determined from Arrhenius analysis, compared to smooth Au@Pt (Au@Pt core-shell arrays without spikes), despite having a similar material composition. The resulting gas-phase photocatalytic performance for our model reaction falls within the upper range of values reported for planar structures^{17,32} and is achieved at a lower optical power density in our study. This cross-study context is qualitative because catalyst identity, loading, reactor geometry, and possible nonlinear intensity-temperature coupling differ across reports. Moreover, wavelength-dependent photocatalytic experiments are consistent with an enhancement mediated by the engineered optical resonance and align with quantum-based calculations that predict increased interfacial carrier populations. Together, our results demonstrate a strategy to decouple photocatalyst performance from intrinsic material properties by setting the optical resonance via the metasurface architecture and provide design principles for planar plasmonic architectures geared toward chemical transformations.

For photocatalytic applications, strong light absorption is essential for increasing the number of photons available for photocatalysis.³⁵ A multilayer metal-insulator-metal (MIM) “perfect absorber” is a metasurface architecture in which a periodic array of subwavelength resonators sits above a dielectric spacer and a reflective backplane to support gap-plasmon modes.^{48,49} In this work, we designed Au NP arrays on this MIM structure consisting of hemispherical Au seeds atop a TiO_2 spacer and an optically thick Au backplane (Figure 1), to achieve strong and spectrally tunable absorption via the gap-plasmon mode in the visible spectrum.⁵⁰ We selected a 20 nm TiO_2 spacer to place the gap-plasmon absorption maximum near 660 nm (Supplementary Figure 4), which maximizes photon harvesting under our illumination conditions. The Au seeds (180 nm diameter, 600 nm pitch, 50 nm height) were fabricated on the $\text{TiO}_2/\text{Au}/\text{Al}_2\text{O}_3$ substrates using SANE and RIE (Figure 1b; see Methods in Supporting Information) with 1 cm $\times$ 1 cm large-scale array size (see details in Supporting Information). The 600 nm lattice pitch comes from our SANE mask set for robust large-area patterning; at normal incidence with air as the superstrate, this period does not exhibit narrow surface-lattice resonances, and the optical response is dominated by a localized gap-plasmon mode in each unit cell.

Spiky Au NPs were synthesized by chemically growing Au protrusions on the Au seed sample.⁴⁶ Prior to synthesis, an annealing process was performed to enhance the surface crystallinity of the template seeds, which proved critical for achieving uniform seed-mediated growth on the substrate.⁴⁶ The growth process employed a Good's buffer containing EPPS (*N*-(2-hydroxyethyl)piperazine-*N'*-3-propanesulfonic acid), which is known to promote anisotropic deposition on {111}-type crystal facets.⁴⁷ EPPS acts both as a reducing and shape-directing agent: its tertiary amines generate cationic radicals to reduce gold ions, while its hydroxyl and propane-sulfonate groups facilitate anisotropic, site-specific Au deposition.⁴⁷ Using fixed concentrations of HAuCl_4 (0.2 mM) and EPPS (400 mM), the reaction over 30 min resulted in spiky Au NPs with elongated spikes approximately 30 nm in length, featuring sharp, high positive-curvature tips with diameters ranging from 5 to 10 nm (Figure 1c).

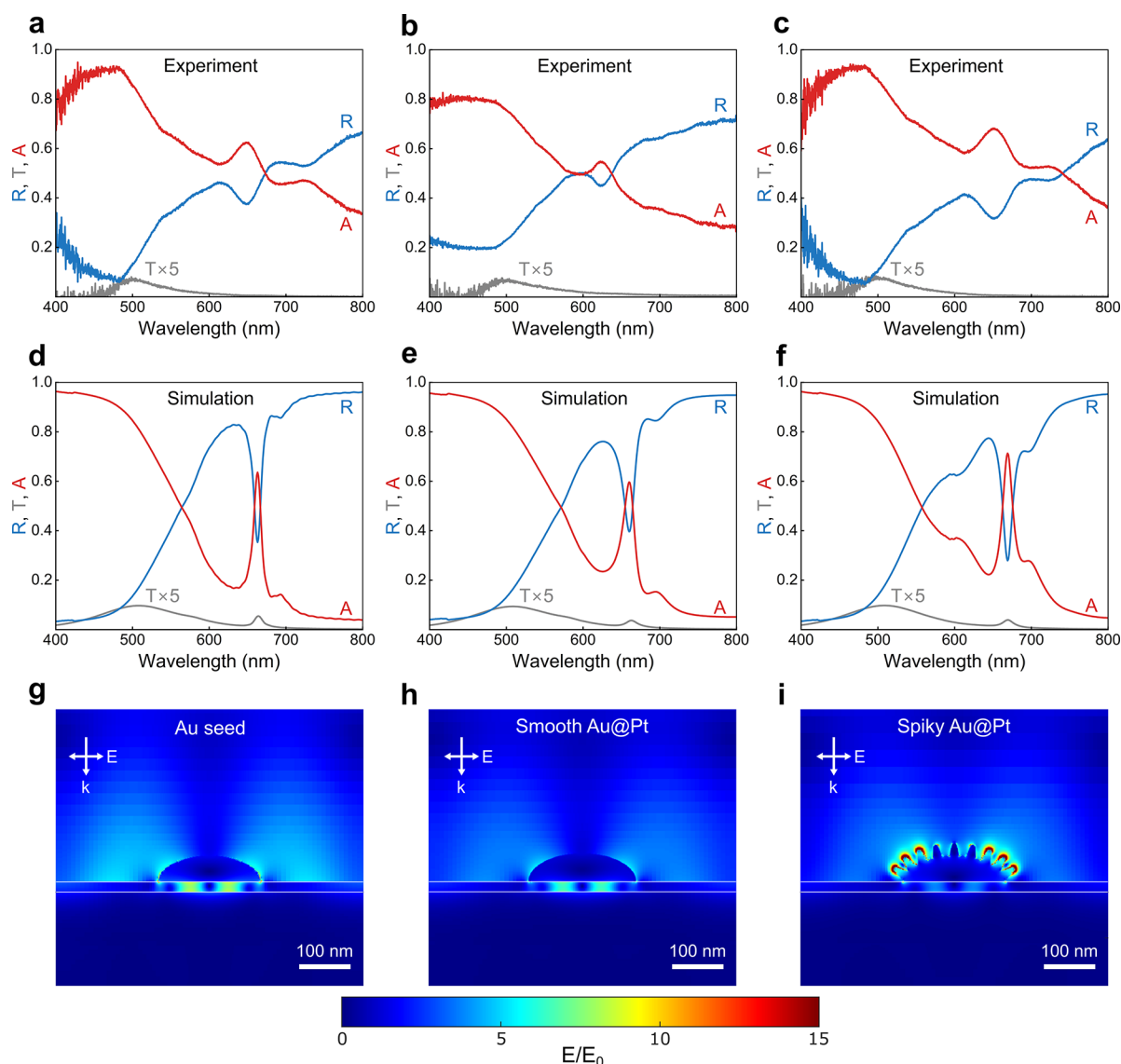


Figure 2. Optical properties of AuPt bimetallic plasmonic lattices. (a–c) Measured reflectance, transmission, and absorption spectra of (a) Au seed array, (b) smooth Au@Pt, and (c) spiky Au@Pt. (d–f) Simulated reflectance, transmission, and absorption spectra of (d) Au seed array, (e) smooth Au@Pt, and (f) spiky Au@Pt. (g–i) Simulated electric-field profiles of each unit cell in air at resonance: Au seed array ($\lambda = 663$ nm), (h) smooth Au@Pt ($\lambda = 661$ nm), and (i) spiky Au@Pt ($\lambda = 669$ nm). Pt is omitted in panel (i) for clarity; Pt-included maps in [Supplementary Note 9](#).

Using H_2PtCl_6 as the metal precursor, Pt was sequentially grown on the spiky Au NPs via a seed-mediated approach. Due to the lower reduction potential of Pt ($E^\circ = 0.68$ V) compared to Au, L-ascorbic acid was used as the reducing agent at an elevated temperature of 90 °C. After 15 min, the growth solution containing the substrate turned from transparent yellow to dark gray, confirming the reduction of Pt^{4+} to metallic Pt NPs ([Supplementary Note 6](#)). This process produced spiky Au@Pt arrays with Pt islands with radii on average of 10–15 nm ([Figure 1d](#)). As a control, we prepared smooth Au@Pt core–shell arrays by applying the same Pt-growth protocol directly to the Au seeds, achieving similar near-surface Pt loading ([Supplementary Notes 2 and 6](#)).

The optical properties of the fabricated bimetallic plasmonic lattices for antenna-reactor photocatalysis were directly measured using an inverted microscope at normal incidence (see [Methods in the Supporting Information](#)). We obtained the reflectance (R) and transmission (T) spectra of all three samples, from which we calculate absorption as $A = 1 - R - T$

([Figure 2a–c](#)). For the bare Au seed ([Figure 2a](#)), the MIM geometry is designed as a resonant absorber, trapping light between the top Au NP and the Au backplane to excite a gap plasmon mode in the TiO_2 spacer. Under practical experimental constraints (e.g., fabrication imperfections and material losses), the absorption reaches a prominent, but sub-unity peak ($A = 0.62$) near 650 nm. When a conformal Pt shell was added (smooth Au@Pt, [Figure 2b](#)), the peak absorption decreased by about 12% and blue-shifted by about 25 nm. The decrease is due to Pt's higher imaginary permittivity⁵¹ introducing additional ohmic losses and weakening the gap-plasmon field. The spiky Au@Pt ([Figure 2c](#)) not only recovers but slightly exceeds the absorption of the Au seed at resonance, with $A = 0.68$ achieved at the designed resonance. The elongated Au protrusions concentrate the near-field via the lightning-rod effect,⁵² enhancing the local density of optical states, which both intensifies coupling into the plasmon mode and boosts scattering cross-section, enhancing overall absorbed photon energy. Below 480 nm, all samples exhibit high

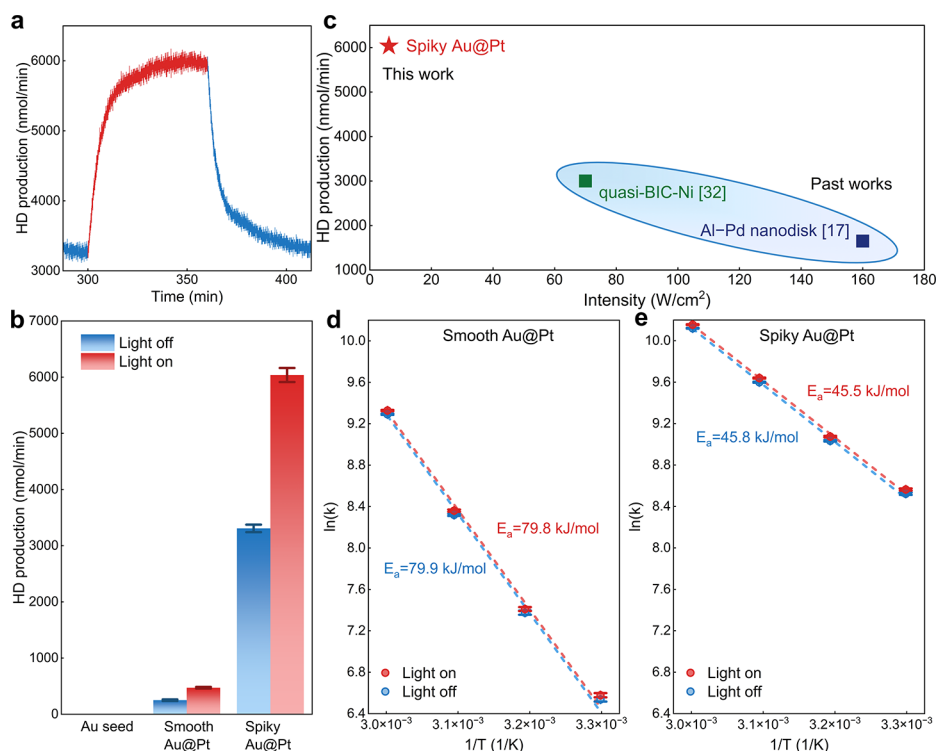


Figure 3. Photocatalytic test of the plasmonic lattices. (a) An example of the mass spectrometer trace of HD production is plotted to show the light on (red line trace) and light off (blue line trace) as a function of time for spiky Au@Pt. (b) HD production for each fabricated structure for H₂-D₂ exchange reaction performed at room temperature with white light (400 to 800 nm) on and off experiments. (c) The same H₂-D₂ reaction is compared among planar structures at room temperature operation. Compared to previous works (quasi-BIC-Ni system³² and Al-Pd nanodisk system¹⁷), our spiky Au@Pt structures exhibit a higher HD production rate while using lower light intensity. This cross-study context is qualitative, given differences in catalyst identity, loading, and reactor geometry. Reproduced from ref 17. Copyright 2016 American Chemical Society. Reproduced from ref 32. Copyright 2024 American Chemical Society. (d) Arrhenius plot for smooth Au@Pt with white light (400 to 800 nm) on and off. (e) Arrhenius plot for spiky Au@Pt with white light (400 to 800 nm) and off. All white light illumination is kept at 500 mW power on a beam spot with a Gaussian diameter of 3.2 mm (intensity is 6.21 W/cm²).

absorption due to Au interband transitions. Measured and simulated spectra of the unpatterned TiO₂/Au/Al₂O₃ substrate (control) are provided in the Supporting Information and do not exhibit these resonant features, indicating that the observed response originates from the patterned arrays.

To validate these observations, we performed full-wave electromagnetic simulations using the measured geometries from the fabricated samples (Methods in Supporting Information). The simulated *R*, *T*, and *A* (Figure 2d–f) closely mirror the experimental trends: the bare Au seed shows the highest simulated absorption (Figure 2d) at the gap plasmon resonance, smooth Au@Pt suffers a dip in absorption (Figure 2e), and spiky Au@Pt exceeds the Au seed by around 13% in peak absorption with a narrow line width (Figure 2f). Subtle shoulders adjacent to the main peak in both measured and simulated spectra correspond to first-order diffractive modes arising from the 600 nm periodicity, indicating radiative coupling between neighboring Au NPs.⁵³

To illustrate the near-field enhancement mechanism, we present simulated electric-field profiles of each unit cell in air at each resonance (Figure 2g–i; see Methods in the Supporting Information). In the bare Au seed array (Figure 2g), a strong gap-plasmon mode concentrates electric fields in the TiO₂ spacer between the top Au NP and the bottom Au backplane, accompanied by a dipolar localized surface plasmon resonance around the Au NP. TiO₂ serves as a high-index spacer whose thickness sets the gap-plasmon resonance and field confinement (see Supporting Information for details on thickness

effects). Introducing a conformal Pt shell (Figure 2h) largely preserves this gap mode but slightly reduces its intensity within the TiO₂ layer, consistent with the decreased absorption observed theoretically and experimentally. Adding protrusions around the Au seed enhances the electric fields at the sharp tip regions (Figure 2i). In the spiky Au@Pt, the spacer-tuned gap mode hybridizes with tip resonances, redistributing part of the energy from the TiO₂ spacer into tip-localized hotspots on the top particle. The prominent peak near ~650 nm therefore corresponds to a gap-tip hybrid: the gap mode sets the spectral position, while the spikes amplify surface fields that coincide with Pt and are most relevant for photocatalysis. Beyond the present design, recent work has demonstrated that the density and length of Au protrusions can be systematically tuned by adjusting growth conditions such as solution pH and buffer concentration, offering additional routes to optimize plasmonic response.⁴⁶ These localized hotspots reach nearly an order of magnitude higher field enhancement compared to that of the bare seed structure, although the gap-plasmon mode in the spacer is modestly attenuated. Such intense, resonance-mediated energy concentration in Au, together with spike-induced localization, enhances delivery of energy to nearby Pt domains (see Supplementary Note 9), providing a plausible plasmonic field-assisted pathway for the enhanced activity described below. In our system, spike-tip localization is the dominant contributor, the MIM stack sets the spectral position and increases absorption, and the gap field in the TiO₂ spacer

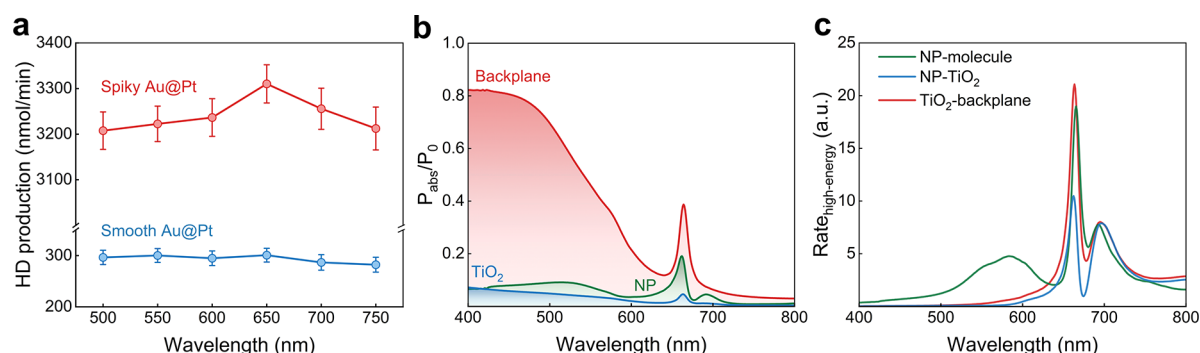


Figure 4. Insights of bimetallic plasmonic lattice for photocatalysis. (a) Wavelength dependence of HD production for smooth Au@Pt and spiky Au@Pt at room temperature. Excitation was provided by a supercontinuum laser with a tunable center wavelength using a 50 nm bandwidth filter and 100 mW power on a beam spot with a Gaussian diameter of 2.5 mm (intensity is 2.04 W/cm²). (b) Calculated ratio between the absorbed power and the incident photon energy for each structure as a function of wavelength (Au seed NP, TiO₂ spacer, and Au backplane). (c) Calculated high-energy carrier generation rate for each interface as a function of wavelength (top Au NP-gas molecule, top Au NP-TiO₂ spacer, and TiO₂ spacer-Au backplane).

mainly funnels energy toward the tips and is attenuated within the spacer.

Having established the optical properties of the bimetallic plasmonic lattices, we next evaluated their photocatalytic performance. We selected the H₂-D₂ exchange reaction ($H_2 + D_2 \rightleftharpoons 2HD$) as a probe reaction, especially as a benchmark reaction for planar structures employing antenna-reactor designs.^{17,32} The HD product was monitored in real time using an in-line mass spectrometer positioned downstream of the photoreactor (Methods in the Supporting Information). An example of the real-time HD ($m/z = 3$) trace for the spiky Au@Pt at room temperature is presented in Figure 3a, clearly demonstrating an enhancement of the HD formation upon white light illumination compared to the dark condition, where thermal catalysis dominates. Such a photocatalytic enhancement is observed for both the spiky Au@Pt and smooth Au@Pt samples (Figure 3b). In contrast, the bare Au seed sample, which lacks active Pt sites, exhibited no detectable HD production under either dark (thermal) or illuminated (photocatalytic) conditions. As an additional control, spiky Au samples without Pt show no detectable HD formation under identical conditions, comparable to the Au seed control (Supplementary Note 7), supporting the idea that Pt domains are required and that Au spikes alone do not catalyze the isotopic exchange. We observed that spiky Au@Pt exhibited a photocatalytic rate approximately 13-fold higher than that of the smooth Au@Pt control sample, despite using an identical Pt-growth protocol and exhibiting comparable near-surface Pt loading (Supplementary Note 6). This enhancement suggests complementary contributions from the sharp-tipped anisotropic Au structures and the antenna-reactor design. Furthermore, under our flow reactor conditions, the measured rate exceeds values previously reported for planar photocatalysts,^{17,32} while being achieved at lower illumination intensity in our study (Figure 3c). This cross-study comparison is qualitative because catalyst identity, loading, and reactor geometry differ across reports. Normalizing the reaction rates to active Pt surface sites (i.e., turnover frequency, TOF) could provide further insight into the intrinsic catalytic efficiency. However, accurately determining the exposed active sites in our planar structures remains experimentally challenging because of the intrinsically low surface area and the limitations of conventional titration approaches. While techniques such as temperature-programmed desorption (TPD) could in principle

provide additional information, such measurements over an extended temperature range are beyond the scope of the present work. Previous benchmark studies on planar antenna-reactor photocatalysts also reported performance without such normalization.^{17,32} Therefore, consistent with established literature, we compare performance based on the measured reaction rate from samples with comparable near-surface Pt loading (Supplementary Note 6) and illumination intensity, and we recommend the further development of chemically sensitive analysis for low-surface area catalytic platforms,⁵⁴ such as hydrogen underpotential deposition (HUPD) to estimate the active Pt surface area.⁵⁵

The photocatalytic performance of the bimetallic plasmonic lattices was further examined at different reaction temperatures to quantitatively assess their activation barriers (E_a) using Arrhenius analysis. Temperature-dependent H₂-D₂ exchange experiments were conducted over the temperature range from 30 to 60 °C at increments of 10 °C for both smooth Au@Pt and spiky Au@Pt samples using a PID-controlled heating stage (Methods in Supporting Information). At each temperature set point, the sample was stabilized under illumination, and the PID controller compensated for the photothermal load (predicted $\Delta T_0 \approx 1.5$ K from analytic expressions for periodic NP arrays under Gaussian illumination,^{56,57} Supplementary Note 8). For smooth Au@Pt (Figure 3d) and spiky Au@Pt (Figure 3e), the Arrhenius slopes under illuminated and dark conditions are similar within the experimental uncertainty. Accordingly, over this temperature window we do not observe an illumination-induced change in the apparent activation barrier. Importantly, under identical illumination, the spiky Au@Pt (Figure 3e) exhibits a lower apparent activation barrier (the activation energy extracted from Arrhenius fits to measured rates), approximately half that of the smooth Au@Pt control (Figure 3d). We attribute this performance difference primarily to the anisotropic geometry, which concentrates electromagnetic energy at sharp Au tips, thereby promoting plasmonic-field-assisted reactivity. Thus, the spiky Au@Pt architecture improves reaction rates and lowers the apparent barrier relative to the smooth Au@Pt control, making this highly anisotropic antenna-reactor design promising for light-driven chemical transformations.

To elucidate the underlying mechanisms driving the enhanced photocatalytic performance observed for our bimetallic plasmonic lattices, we conducted a wavelength-

dependent H₂-D₂ exchange reaction at room temperature. Employing a supercontinuum laser source coupled with a variable-bandwidth-tunable filter (Methods in Supporting Information), we evaluated the wavelength-specific photocatalytic response over a 50 nm bandwidth for both spiky Au@Pt and smooth Au@Pt (Figure 4a). Across the entire wavelength range investigated, the spiky Au@Pt structure consistently exhibited approximately 1 order of magnitude higher photocatalytic rate compared to the smooth Au@Pt control sample. This difference highlights that introducing highly anisotropic spiky Au geometries, compared to smooth Au NPs, enhances catalytic activity beyond their similar far-field optical absorption (Figure 2). More importantly, the photocatalytic rate of the spiky Au@Pt exhibits a wavelength dependence, with maximum HD production at 650 nm, which coincides with the experimentally observed plasmon resonance shown in the optical measurements (Figure 2). This spectral alignment strongly suggests the central role of plasmon resonance-mediated near-field enhancement on the observed chemical transformations.

To elucidate the observed wavelength-dependent photocatalytic enhancement, especially the prominent resonance at approximately 650 nm, we conducted numerical electromagnetic simulations and quantum-based calculations. First, we quantitatively analyzed the spectral dependence of energy absorption in each component of the plasmonic structure using finite-difference time-domain (FDTD) simulations, specifically in the top Au NP, the TiO₂ spacer, and the Au backplane. For simplicity, Pt absorption was neglected in this calculation due to its relatively minor optical contribution in the wavelength range of interest. The absorbed power density (P_{abs}), normalized to the incident photon flux (P_0), was calculated according to

$$P_{\text{abs}} = -0.5\omega |E|^2 \epsilon_0 \text{Im}(\epsilon_r)$$

where ω is the frequency, E is the local electric field, ϵ_0 is the vacuum permittivity, and $\text{Im}(\epsilon_r)$ is the imaginary part of the local relative permittivity. Absorption by the Au backplane dominates across the visible spectrum, particularly at shorter wavelengths (Figure 4b), due to strong interband transitions. However, a distinct peak centered around 650 nm is visible for all of the structures, especially for the Au NP and the TiO₂ spacer layer as the highest absorption. These results suggest strong resonant coupling and gap-plasmon excitation at this wavelength, which likely drives the observed photocatalytic response. Additionally, the dominance of the Au backplane absorption underscores the contribution of photothermal effects across the entire visible range, complementing plasmonic-field assisted chemistry.

Subsequently, we analyzed hot-carrier generation across the plasmonic interfaces using a quantum-mechanical surface-based model that accounts for overbarrier carrier excitation.⁵⁸ In this framework, the high-energy carrier generation rate at metal interfaces is given by

$$\text{Rate}_{\text{high-energy}} = \frac{2}{\pi^2} \frac{e^2 E_F^2}{\hbar} \frac{(\hbar\omega - \Delta E_b)}{(\hbar\omega)^4} \int |E_{\text{normal}}|^2 dS$$

where ΔE_b represents the interfacial energy barrier that carriers must overcome (e.g., the Schottky barrier at Au-TiO₂ interface or the molecule energy offset between the metal and adsorbed H₂), E_F is the Fermi level of the metal, and the surface integral is evaluated over all relevant metal interfaces in the structure,

namely the Au NP surface exposed to H₂ (NP-molecule), the Au NP bottom surface contacting the TiO₂ spacer (NP-TiO₂), and the Au backplane-TiO₂ interface (Figure 4c). While this model captures general trends, it does not account for all interfacial complexities in the experimental system. This formulation shows that hot-carrier generation scales with the local normal electric field intensity at the interface. Our calculations reveal that hot-carrier generation is enhanced at a plasmonic resonance wavelength around 650 nm, consistent with both the experimentally observed photocatalytic peak and the simulated optical absorption. Although the geometric interfacial area between the Au NP and surrounding H₂ molecules is smaller than that of the extended TiO₂-Au backplane, the intense near-field localization at plasmonic resonance enhances the surface-integrated hot-carrier generation rate, making it comparable in magnitude to the larger backplane interface.

Collectively, our experimental and theoretical analyses suggest that the enhanced catalytic performance of spiky Au@Pt arises from the combined effects of (i) modest photothermal heating from broadband absorption, (ii) strong near-field enhancement at sharp Au spike tips, and (iii) resonance-mediated hot-carrier generation localized near catalytic Pt sites. Although we cannot yet apportion their relative weights, the data indicate that spike-induced localization in Au and resonance-mediated energy concentration, which deliver energy to adjacent Pt domains with large imaginary permittivity^{59,60} (Supplementary Note 9), are the primary contributors, whereas photothermal effects are minor under our flux (Supplementary Note 8). These insights provide a rationale for the improvements in both reaction rates and catalytic energetics achieved through antenna-reactor design. More broadly, they point to general design principles for photocatalytic metasurfaces: maximizing nanoscale morphological features (e.g., spike density) to concentrate fields, positioning catalytic sites (e.g., Pt) at plasmonic hot spots, and tuning the underlying metasurface (e.g., TiO₂ spacer thickness) to drive absorption closer to unity.

In summary, we have demonstrated bimetallic plasmonic lattices with morphology-engineered spiky Au NPs integrated with catalytic Pt sites (spiky Au@Pt), achieving improvements in photocatalytic performance. Compared to the smooth Au@Pt control, the spiky structures show photocatalytic rates approximately 13 times higher and apparent activation energies nearly 50% lower for H₂-D₂ exchange, exceeding rates previously reported planar photocatalysts while operating under lower illumination intensities.^{17,32} Through combined optical characterization, wavelength-dependent catalytic studies, and theoretical modeling, we identified key optical and interfacial effects that underlie the improved performance. These results not only advance the understanding of plasmon-mediated chemical transformations but also outline design considerations for future planar photocatalysts. Our fabrication strategy and demonstrated catalytic performance highlight the potential of large-area, low-cost plasmonic reactors for light-driven chemical processes.

■ ASSOCIATED CONTENT

Data Availability Statement

The data supporting the findings of this study are included within the article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.nanolett.5c04789>.

Methods; smooth Au@Pt lattice; optical image of the fabricated plasmonic bimetallic nanoparticle lattices; optical properties of the substrate-only; effect of spacer layer thickness; XPS survey scans of smooth Au@Pt and spiky Au@Pt; spiky Au (no Pt) control; photothermal temperature rise under broadband illumination; near-field comparison at resonance and off resonance (PDF)

AUTHOR INFORMATION

Corresponding Authors

Dayne F. Swearer – Department of Chemical and Biological Engineering, Northwestern University, Evanston, Illinois 60208, United States; Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0003-0274-4815; Email: dayne@northwestern.edu

Teri W. Odom – Department of Chemistry and Department of Materials Science and Engineering, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0002-8490-292X; Email: todom@northwestern.edu

Authors

Guanyu Lu – Department of Chemical and Biological Engineering, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0001-8960-0464

Nam Heon Cho – Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0002-0316-2422

Sachin P. Kulkarni – Department of Electrical and Computer Engineering, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0003-3740-3527

Matthew Hershey – Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0009-0006-7198-5814

Xin Wen – Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States

Sang-Min Park – Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0002-4306-0375

Koray Aydin – Department of Electrical and Computer Engineering, Northwestern University, Evanston, Illinois 60208, United States; orcid.org/0000-0002-3268-2216

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acs.nanolett.5c04789>

Author Contributions

G.L. performed the photocatalytic experiments, electromagnetic simulations, and data analysis. G.L. and S.P.K. performed the optical measurements. N.H.C. and S.M.P. fabricated Au seed arrays. N.H.C. and X.W. synthesized spiky Au@Pt, and N.H.C. provided the writeup for the synthesis details. M.H. performed XPS survey scans on smooth and spiky Au@Pt. G.L. wrote the manuscript with input from all other authors. K.A., T.W.O., and D.F.S. acquired the funding and supervised the work.

Notes

The views and conclusions contained in this document are those of the authors and should not be interpreted as representing the official policies, either expressed or implied, of the U.S. Army or the U.S. Government. The authors declare no competing financial interest.

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